جامعة الإمام عبد الرحمن بن فيصل
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY

Logistics Paperwork Process Automation System (Web-Based Document Upload, Data Extraction, and Form Generation)

Course

BIS405 – Graduation Project

Area

Digital Systems and Solution Design

Team Name:

Team Members & IDs:

[Norah Alshehri] [2230004682]
[Bayader Alghamdi] [2230003715]
[Shaykhah Mohsen] [2230005764]
[maha Alnafea] [2230003005]

Supervisor: Dr. linah Saraireh

Date: 2/11/2026

1. Problem and Value

1.1 Problem Statement

In B2B logistics operations, staff often enter shipment and customer data hand from documents such as Invoices, Packing Lists, and AWB/BOL into internal forms and operational systems. This hand work raises processing time, adds data-entry errors, and makes document tracking and auditing hard (e.g., absent fields, repeat entries, uneven formatting).

1.2 Users / Stakeholders and Value

Operations staff (shipment processing) will benefit because they can handle shipment cases faster and reduce rework caused by manual entry and missing information. Customs/clearance coordinators (if applicable) will benefit because the system helps keep documents more complete and consistent across cases. Sales/account managers will benefit because they can respond faster to customer requests that depend on accurate paperwork. Management will benefit because dashboards provide clear visibility on shipment volume, delays, and document quality, which supports better decisions.

1.3 Primary KPI(s) and Targets

Table 1: Primary KPI(s) an Targets

Measurement note	KPI	Target	Unit
Targets will be measured against a recorded baseline (time study + sample cases).	**KPI 1:** Processing time per shipment paperwork package	≥ 35% drop vs baseline[1]	Minutes per case
Targets will be measured against a recorded baseline (time study + sample cases).	**KPI 2:** Data-entry error / correction rate	≥ 40% drop vs baseline[2]	Fixes per case
Targets will be measured against a recorded baseline (time study + sample cases).	**KPI 3:** Document completeness rate at first review	≥ 20% gain vs baseline[3]	Cases with no absent critical fields

[1] RPA studies report 40–70% process-time reduction for repetitive tasks.

[2] An IDP study reports document error reduction around 85% with ML-based automation.

[3] IDP guidance highlights using validation rules/confidence routing to improve field-level data quality and consistency (typical 90–98% field-level accuracy on standard documents).

-

2. Scope and Constraints

2.1 In-Scope

- Web application for document upload (PDF/Excel; image/OCR as stretch goal).
- Automated extraction of key fields from standard logistics documents (e.g., invoice number, dates, shipper/consignee, weight, quantity, total value, reference numbers).
- Review-and-approve interface for users to check/edit extracted values.
- Export of standard outputs (e.g., Excel/PDF form or internal template).
  Database storage for cases, extracted fields, document details, and audit trail.
  Basic dashboard for volume and quality KPIs.

2.2 Out-of-Scope

- Full integration with external government customs platforms or ERP systems (only optional future work).
- Full-scale OCR for all document types with high accuracy guarantees (optional stretch goal).
- Automated legal compliance decisions (e.g., HS classification or regulatory approvals).

2.3 Key Assumptions and Constraints

Table 2: Key Assuptions and Constraints

Time	limited to semester schedule.
Data access	Unavailable Sensitive government documentation: official customs declarations
Privacy/ethics	any sensitive identifiers must be removed/anonymized; access is restricted to authorized users.
Technology	the solution should be possible to implement, test, and demo reliably within the course setting.

3. Requirements

3.1 User Stories

US1 – Upload Document Package As an operations user, I want to upload one or more documents for a shipment so that the system can process them. **Acceptance Criteria:**

- User can upload PDF/Excel (and optionally image) files up to [X MB].
- Each upload is tied to a shipment case ID.
- System confirms successful upload and stores details (file name, type, timestamp, user).

-

US2 – Extract Key Fields Automatically As an operations user, I want the system to extract key fields from uploaded documents to reduce hand typing. **Acceptance Criteria:**

- System extracts a defined minimum set of fields (Version 1).
- Extracted fields are displayed in a structured form within [Y seconds] for normal file size.
- System marks absent/low-confidence fields for hand review.

US3 – Review, Edit, and Approve As a reviewer, I want to check and fix extracted values before saving them as final. **Acceptance Criteria:**

- Reviewer can edit any extracted field.
- Approved status is saved with user/time.
- All edits are saved (old value, new value, editor, timestamp).

US4 – Generate Standard Output (Form/Report) As a user, I want to export the approved data into a standard form (Excel/PDF) for operational use. **Acceptance Criteria:**

- Export includes all required fields.
  Output follows a consistent company-style templat
  File can be downloaded and is tied to the case history.

US5 – Search and Retrieve Cases As a user, I want to search past cases to quickly retrieve documents and extracted data. **Acceptance Criteria:**

- Search by case ID, invoice number, customer name, or date range.
- Results show case status and last update time.
- Authorized users can view/download documents.

3.2 Critical Non-Functional Requirements (NFRs)

- Security: authentication + role-based access (Admin/Staff/Viewer); safe file access; basic audit trail.
- Performance: extraction + page rendering within okay time for normal file sizes (target values defined during Week 2 baseline).
- Usability: a user can complete upload → review → export with few steps and clear validation messages.
- Reliability: system handles invalid files safely and never replaces stored originals.

4. Data / Inputs & Governance

4.1 Data Sources and Access

- Primary data source: A curated dataset of logistics and trade document sourced from kaggle (Invoices, Packing Lists, AWB/BOL).
- Fallback: synthetic documents created by the team using realistic templates and dummy data.

4.2 Expected Inputs (high-level schema)

- Case Table: case_id, customer, service_type, created_at, status, created_by
- Document Table: doc_id, case_id, doc_type, file_path, upload_time, uploaded_by
- Extracted Fields Table: field_id, case_id, field_name, extracted_value, confidence/flag, source_doc, approved_value, approved_by, approved_at
- Audit Log: change_id, case_id, field_name, old_value, new_value, changed_by, changed_at

4.3 Data Quality and Governance

- Document variety (different layouts) may reduce extraction accuracy; Version 1 will focus on the most common templates.
- Privacy and consent: only anonymized or synthetic data will be used in submissions and demos; access restricted to authorized roles.

5. Method and Feasibility

5.1 High-Level Architecture (Systems)

Web UI → Backend API → Extraction Engine → Database/Storage → Export Service → Dashboard

- **UI:** upload, extracted-field review, case search, export, KPI dashboard.

  Backend API: authentication, case management, document management, extraction job handling.

  Extraction Engine:

  ○ PDF text extraction for text-based PDFs, ○ Excel parsing for spreadsheets, ○ optional OCR module for scanned images (stretch).
- **Storage:** database for structured fields + safe file storage for originals and outputs.

- **Dashboard:** combined KPIs (time saved estimate, correction rate, volume).

5.2 Why This Approach Fits the Problem / KPIs

- Automation directly targets hand-entry time and correction cycles (KPI 1 and KPI 2).
- Review workflow + audit trail improves control and traceability.
- Gradual delivery is possible using the course sprint plan (working increment each sprint).

5.3 Proposed Tools / Stack

- Frontend: React (or HTML/CSS/Bootstrap)
- Backend: Python (FastAPI/Flask)
- Database: PostgreSQL / MySQL / SQLite (based on hosting constraints)
- Extraction: PDF text extraction + Excel parsing; OCR as optional
- Version control& project tracking: GitHub backlog and sprint board

6. Plan, Milestones and Top Risks

Timeline

- Week 2 (Problem discovery & scoping): confirm users, define KPIs, baseline measurement plan, finalize scope.
- Week 3 (Proposal submission): locked scope + requirements + plan (this document).
- Week 4 (Early feasibility / POC): skeleton web app + upload working + parse one sample type.
- Sprint 1: authentication + CRUD for cases/documents + basic database schema.
- Sprint 2: extraction rules for 1–2 document templates + review/edit UI.
- Midterm checkpoint: working demo of end-to-end flow (upload → extract → review → save).
- Sprint 3: export to Excel/PDF + audit logging.
- Sprint 4: dashboard KPIs + testing (unit/integration/usability) + performance tuning.
- Sprint 5 (Freeze & rehearsal): stability, final demo script, fallback video.
- Final (Presentation & handover): live demo + final docs + handover package.

Risks and Mitigations

Risk: Limited access to real documents / privacy restrictions Mitigation: anonymize early; use synthetic templates; document governance and limits clearly.
Risk: Document layout variety reduces extraction accuracy Mitigation: start with the most common templates; implement a review step; store confidence flags; treat OCR as stretch goal.
Risk: Scope creep (too many document types / features) Mitigation: define Version 1 required fields + minimum document types; backlog priority; lock out-of-scope integrations.

7. References

FastAPI. (n.d.). FastAPI documentation. Retrieved February 11, 2026, from https://fastapi.tiangolo.com/

Meta Platforms, Inc. (n.d.). React documentation. Retrieved February 11, 2026, from https://react.dev/

PostgreSQL Global Development Group. (n.d.). PostgreSQL documentation. Retrieved February 11, 2026, from https://www.postgresql.org/docs/

jsvine. (n.d.). pdfplumber [Computer software]. GitHub. Retrieved February 11, 2026, from https://github.com/jsvine/pdfplumber

Artifex Software, Inc. (n.d.). PyMuPDF documentation. Retrieved February 11, 2026, from https://pymupdf.readthedocs.io/

tesseract-ocr. (n.d.). Tesseract OCR [Computer software]. GitHub. Retrieved February 11, 2026, from https://github.com/tesseract-ocr/tesseract

The pandas development team. (2020). pandas-dev/pandas: Pandas [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.3509134

Gazoni, E., & Clark, C. (n.d.). openpyxl documentation. Retrieved February 11, 2026, from https://openpyxl.readthedocs.io/

Jaume, G., Ekenel, H. K., & Thiran, J.-P. (2019). FUNSD: A dataset for form understanding in noisy scanned documents (arXiv:1905.13538). arXiv. https://arxiv.org/abs/1905.13538

clovaai. (n.d.). CORD: A consolidated receipt dataset for post-OCR parsing [Dataset]. GitHub. Retrieved February 11, 2026, from https://github.com/clovaai/cord

KPI(s) References

Roland Berger. (2018). RPA – Tomorrow's must-have technology: Speed up your business with robotic process automation [Report]. https://www.rolandberger.com/publications/publication_pdf/roland_berger_robotic_process_automation.pdf?v=712824

Kumar, A., & Singh, R. (2025). AI-driven intelligent document processing for healthcare and insurance. International Journal of Science and Research Archive, 14(1), 1063–1077. https://journalijsra.com/sites/default/files/fulltext_pdf/IJSRA-2025-0194.pdf

Nutrient. (n.d.). What is intelligent document processing (IDP)? A complete guide. Retrieved February 11, 2026, from https://www.nutrient.io/blog/what-is-intelligent-document-processing/